

Advanced Manufacturing Technology & Prototyping (MDP494)



Lecture 2_Part 1

Introduction to CNC Milling programming (part 2)

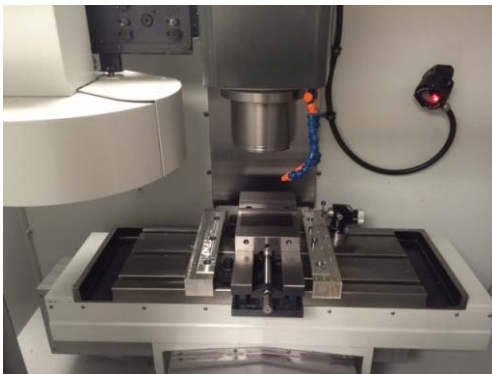
DR. Moustafa Mahmoud

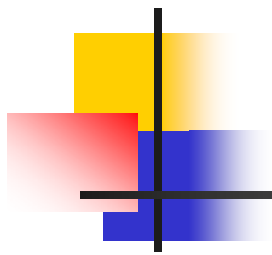
Ph. D. Eng.

Design & Production Dept.,

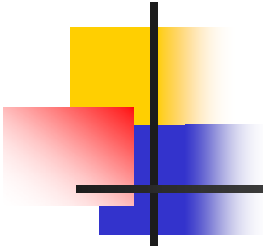
Faculty of Engineering, Ain Shams University

m_m_sayed@eng.asu.edu.eg





2- PROGRAMMING PREPARATORY FUNCTIONS



Feed rate/speed programming, G93 to G97

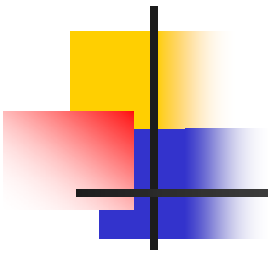
F: Feed rate

- **G93 F:** Feed rate in **1/min** (inverse time feed rate)
- **G94 F:** Constant speed and feed rate in **mm/min**
- **G95 F:** Constant speed and feed rate in **mm/revolution**

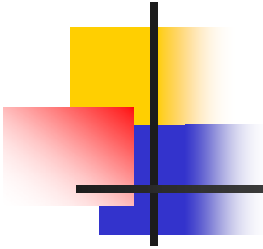
S: Speed

- **G96 S:** Constant cutting velocity in **m/min**
- **G97 S:** Constant cutting velocity in **rev/min**

Ex.: G97 S1000

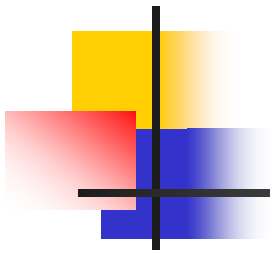


3- PREDEFINED SWITCHING AND AUXILIARY FUNCTIONS

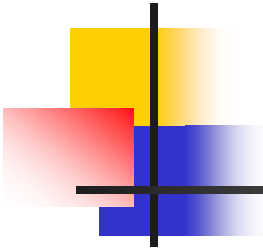


M: Miscellaneous Function

- **M3:** Sense of rotating main spindle **clock wise**.
- **M4:** Sense of rotating main spindle **counter clock wise**.
- **M5:** Spindle **stop** after motion been completed
- **M6:** Tool change
- **M8:** Coolant on
- **M9:** Coolant off
- **M30:** End of program

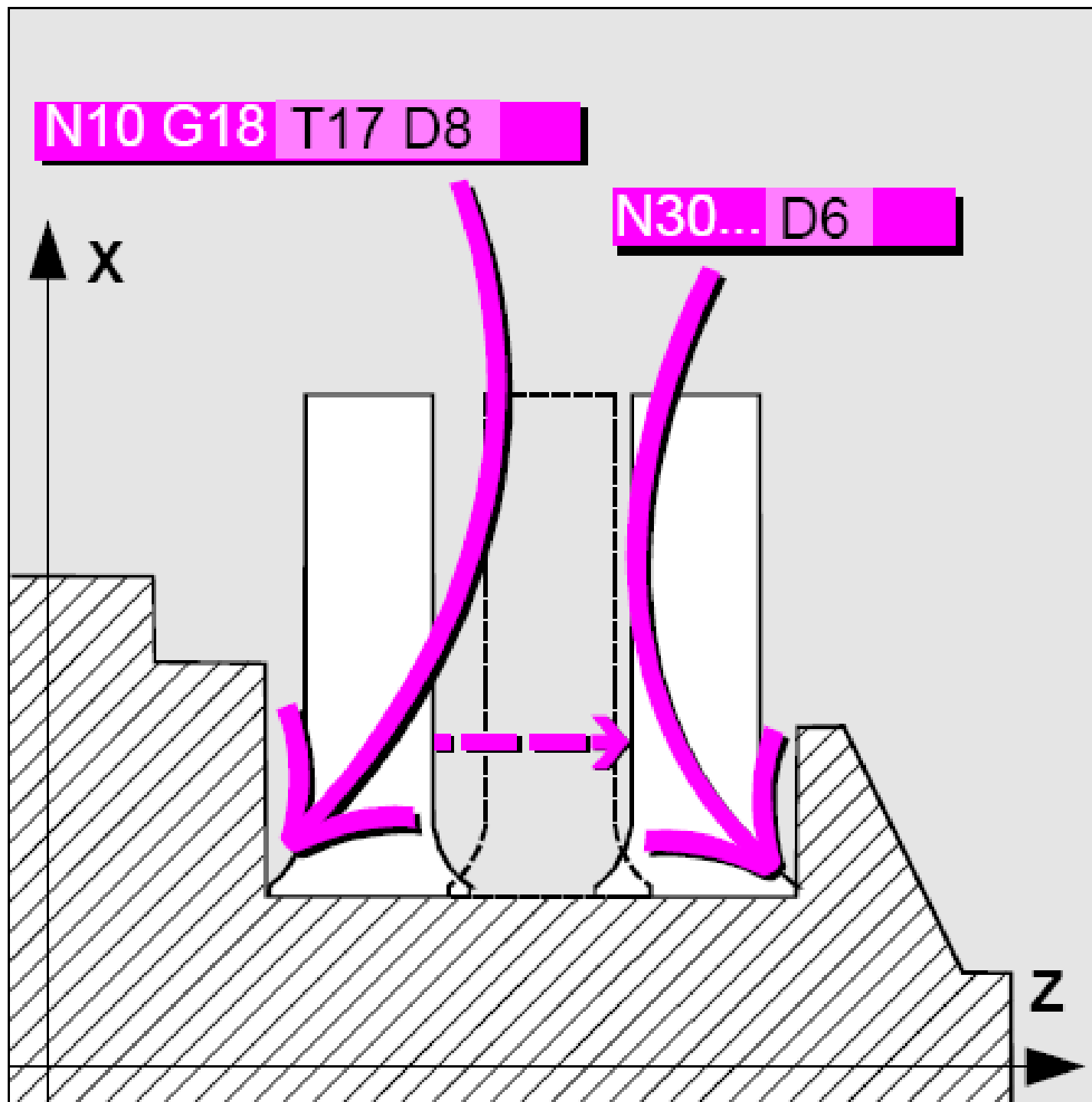
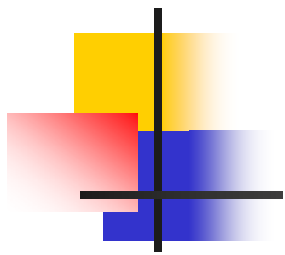


4- TOOL OFFSETS AND COMPENSATIONS

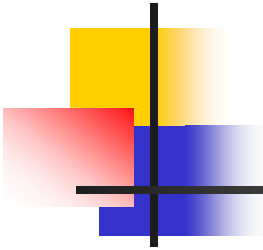


Tool call

- **T...** Call tool number
- **D...** Call tool offset, activate tool length compensation
- The tool length compensation acts automatically after tool offset **D** has been called.

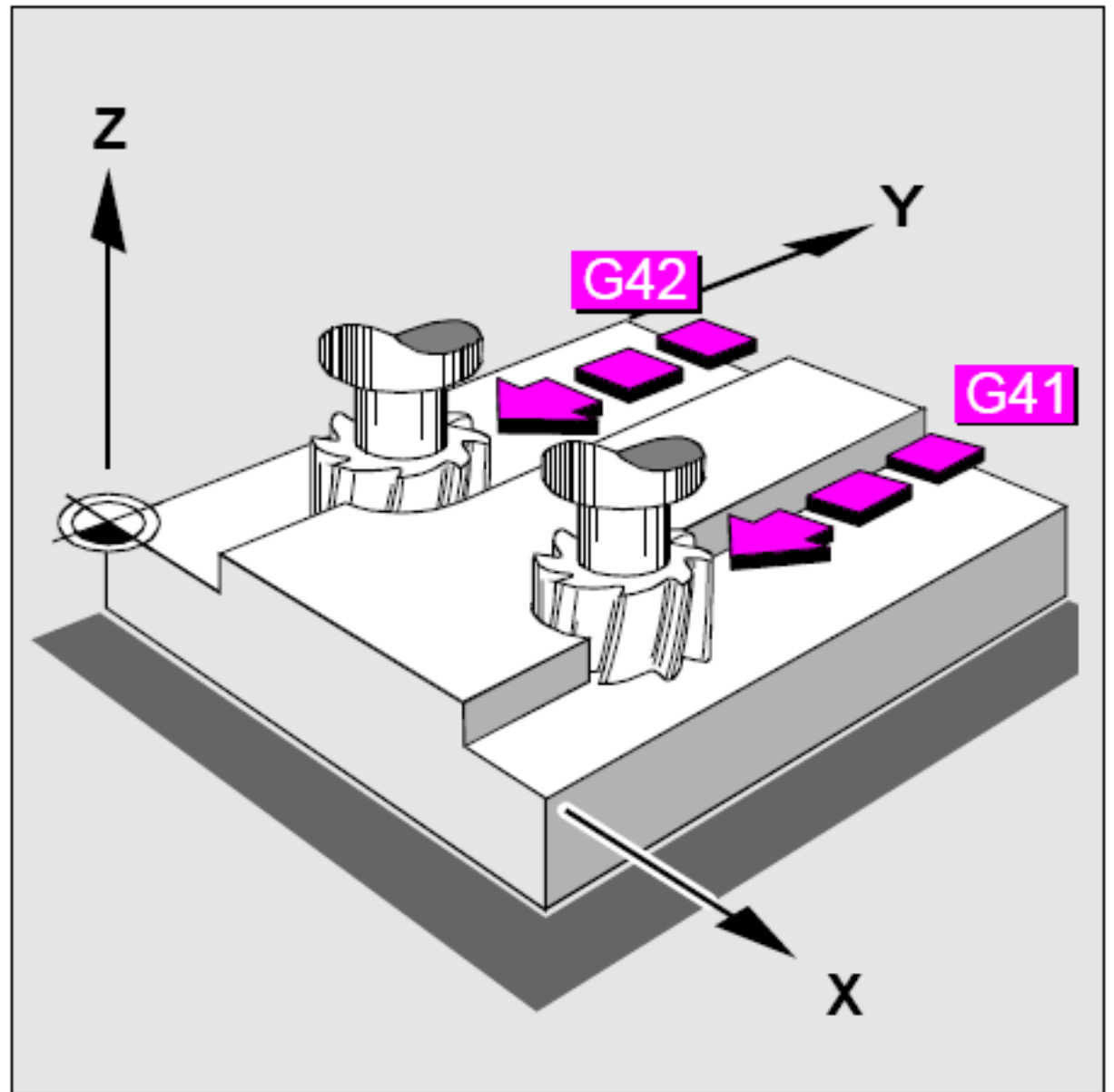


Offset values for left-hand and right-hand tool nose for recessing tool

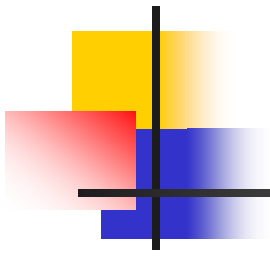


Tool/Cutter radius path compensation, G41/G42/G40

- **G41:** Call for cutter path compensation, tool in traversing direction **left of contour**
- **G42:** Call for cutter path compensation, tool in traversing direction **right of contour**
- **G40:** **Deselection** of cutter path compensation
- The offset acts only in the programmed working plane (G17 to G19).

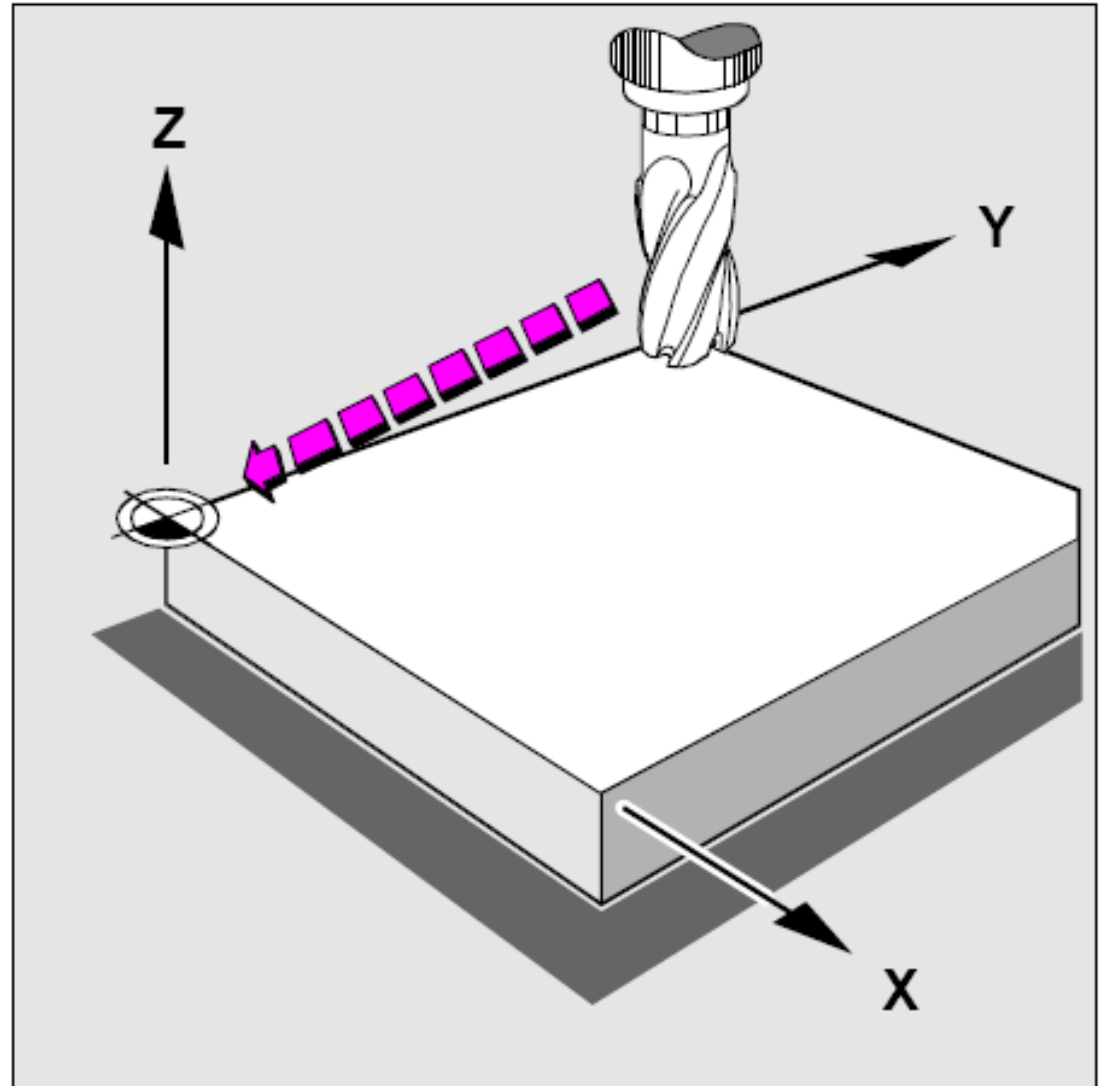


The control computes the tool path



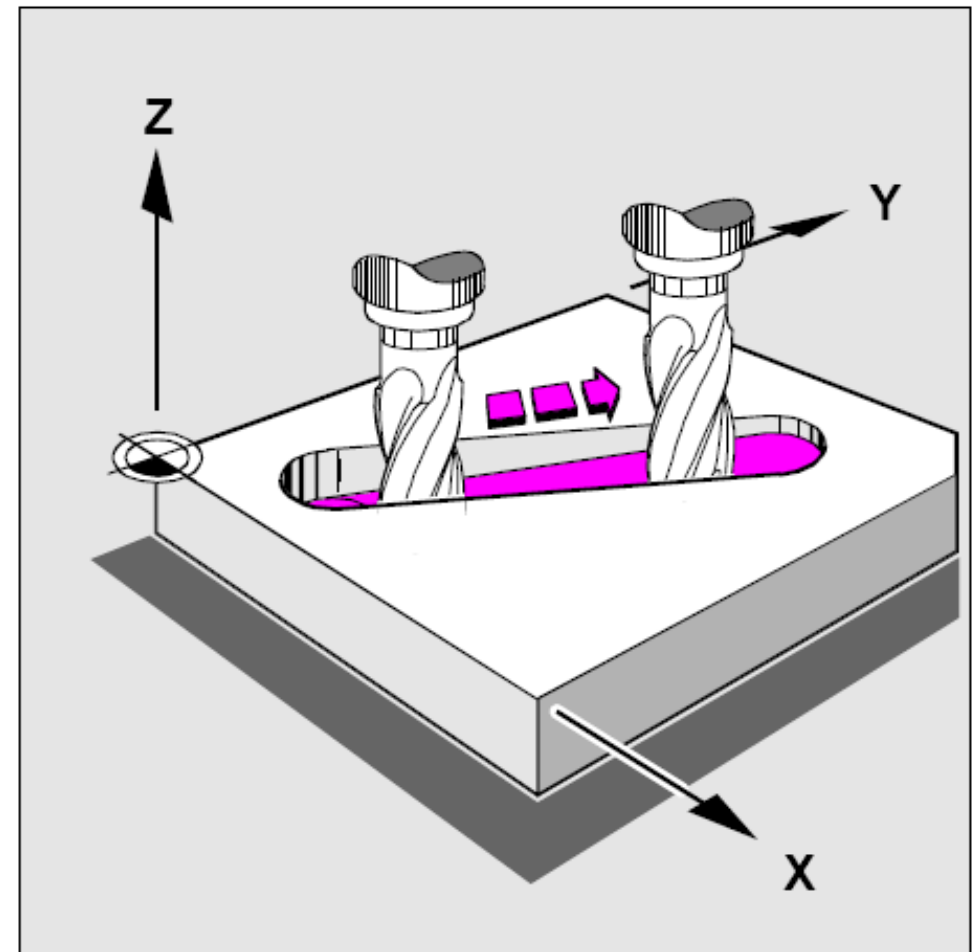
5- PROGRAMMING MOTION COMMANDS

Rapid traverse, G0

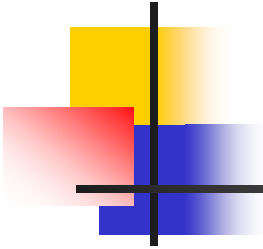


Fast tool positioning in rapid traverse for milling

Linear interpolation, G1



Making an inclined slot



Circular interpolation, **G2/G3**

- With **G2** the tool travels clockwise
- With **G3** counterclockwise.

There are two methods of input data:

1- Using Radius of the circular shape

2- Using Coordinates of the circular shape

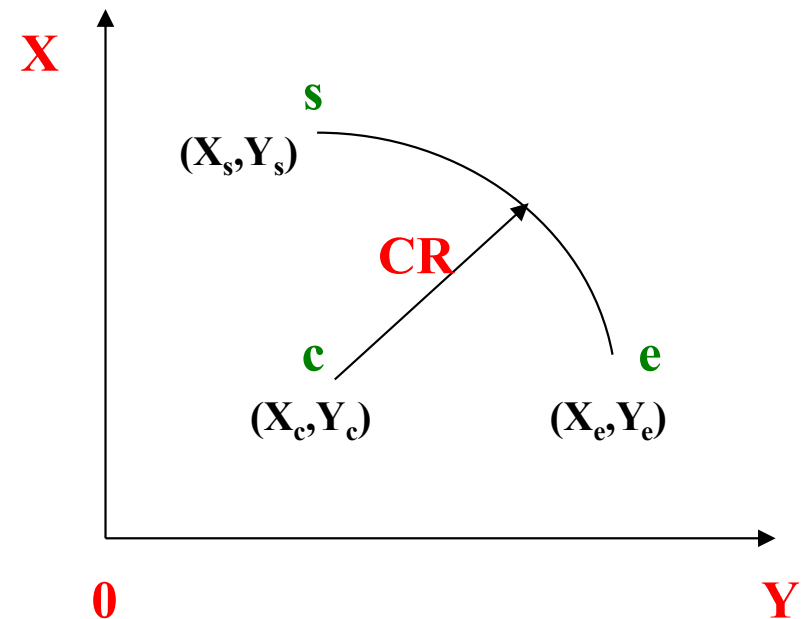
Circular interpolation, G2/G3

1st - Using Radius of the circular shape

The shape of this function:

G2 X 000 Y 000 CR000 F000

Ex. G2 X10 Y30 CR50 F100



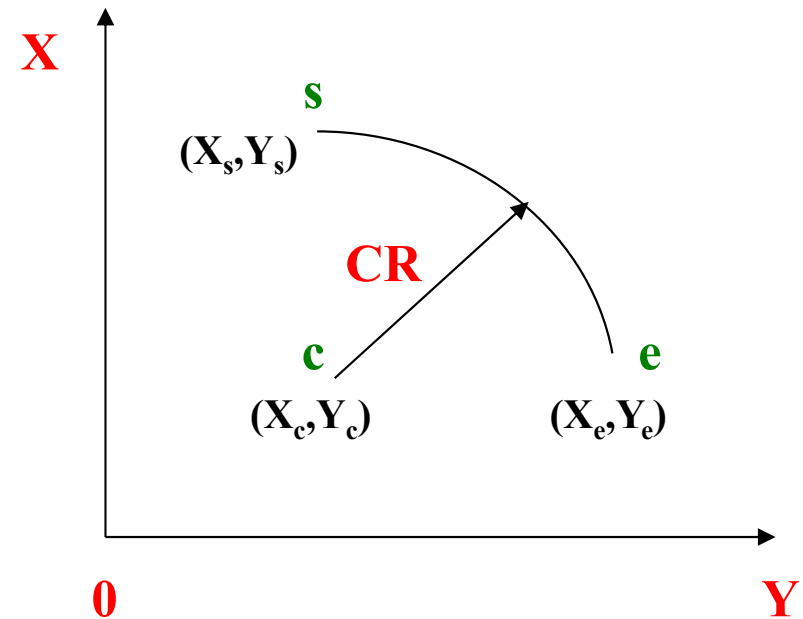
Circular interpolation, G2/G3

2- Using Coordinates of the circular shape

Where:

I: X_{center}

J: Y_{center}

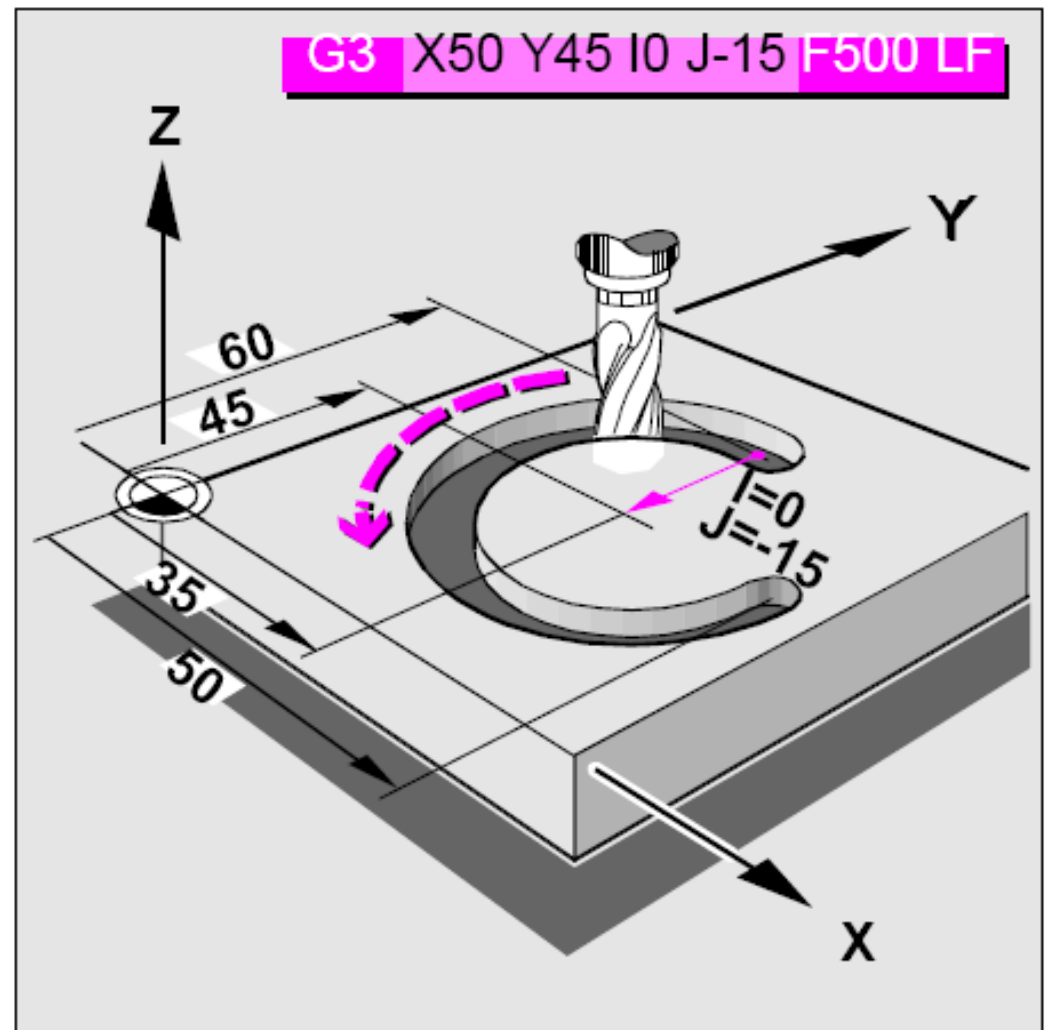


The shape of this function:

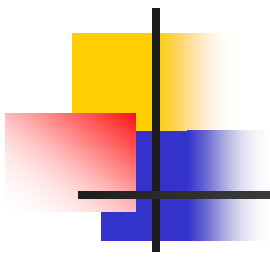
G3 X 000 Y 000 I000 J000 F000

Ex. G2 X50 Y45 I0 J-15 F500

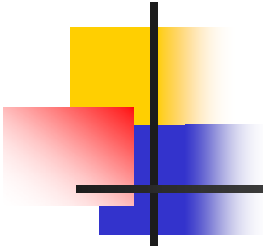
Circular interpolation, G2/G3



Making a circular slot

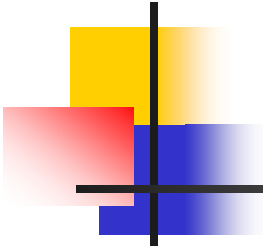


MOST USED **G – CODES** IN PROGRAMMING



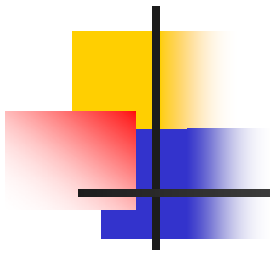
G - Codes

G0	Rapid traverse
G1	Machining traverse
G2	Circular interpolation clockwise
G3	Circular interpolation counterclockwise
G4	Dwell time
G17	Working plane selection XY
G18	Working plane selection XZ
G19	Working plane selection YZ
G40	Cutter radius compensation OFF
G41	Cutter radius compensation ON Left
G42	Cutter radius compensation On Right

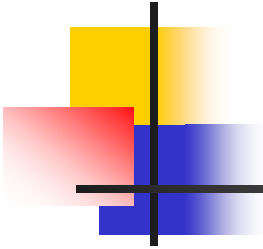


G – Codes, cont.

G53	Deselecting settable zero offset
G54-G57	Settable zero offset
G70	Input system inch
G71	Input system metric
G90	Absolute dimensioning
G91	Incremental dimensioning
G94	Feed in mm/min, inch/min
G95	Rotational feed rate in mm/rev, inch/rev
G96	speed (m/min)
G97	speed (rev/min)

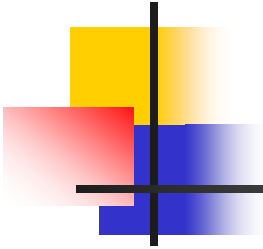


MOST USED **M** – **CODES** IN PROGRAMMING



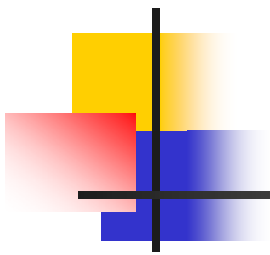
M - Codes

M0	Programmed stop
M1	Optional stop
M2	End of program
M3	Spindle on clockwise
M4	Spindle on counterclockwise
M5	Spindle stop



M - Codes

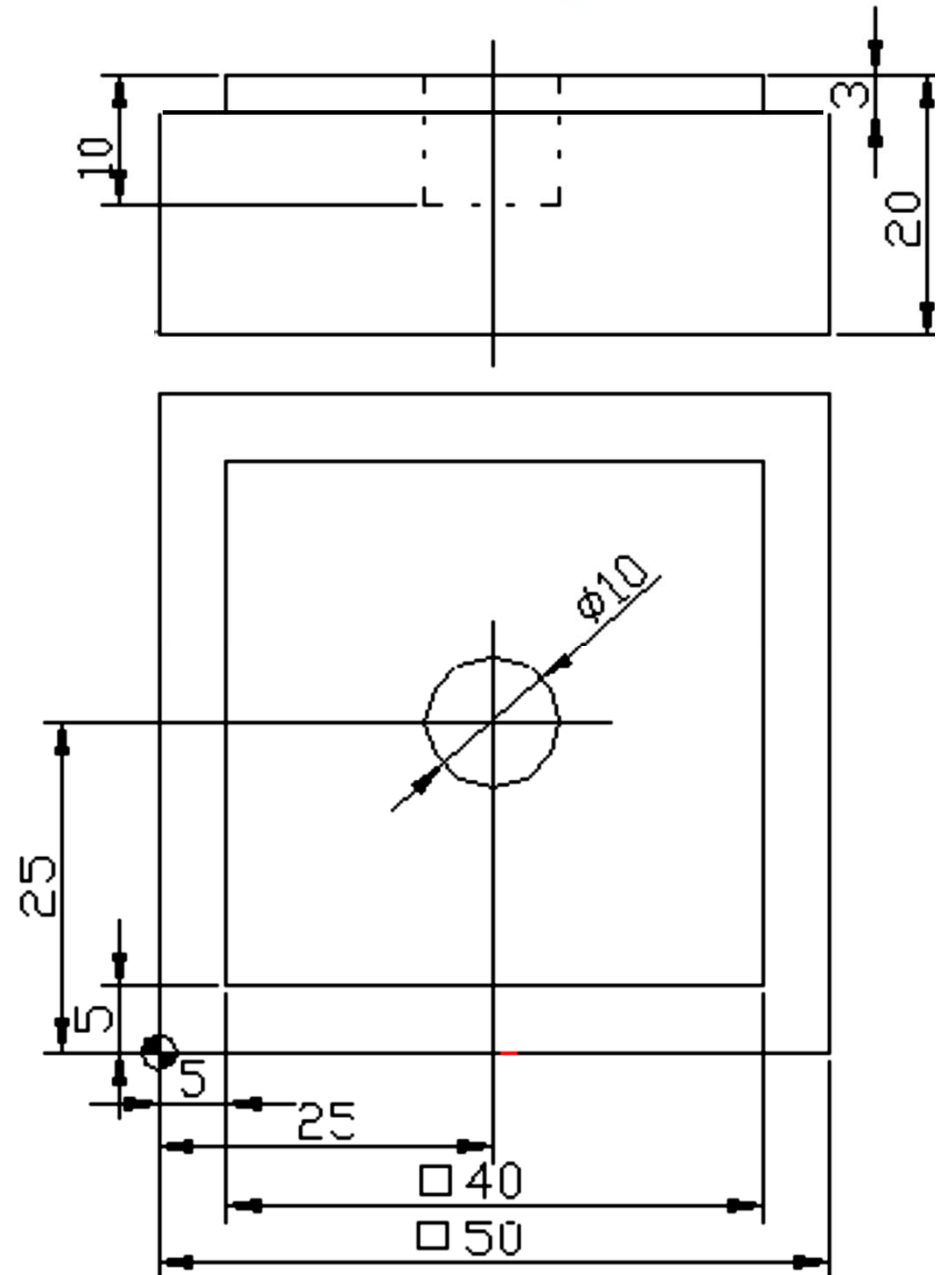
M6	Tool change
M8	Coolant on
M9	Coolant off
M17	Subprogram end
M30	Program end

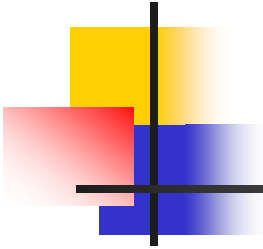


SOLVED EXAMPLE 1

Milling Programming

- Write down the program that may be used to produce the given work piece on a CNC milling machine using the Sinumeric 840D coding.
- Drawing of the work piece is shown and all dimensions are in mm.





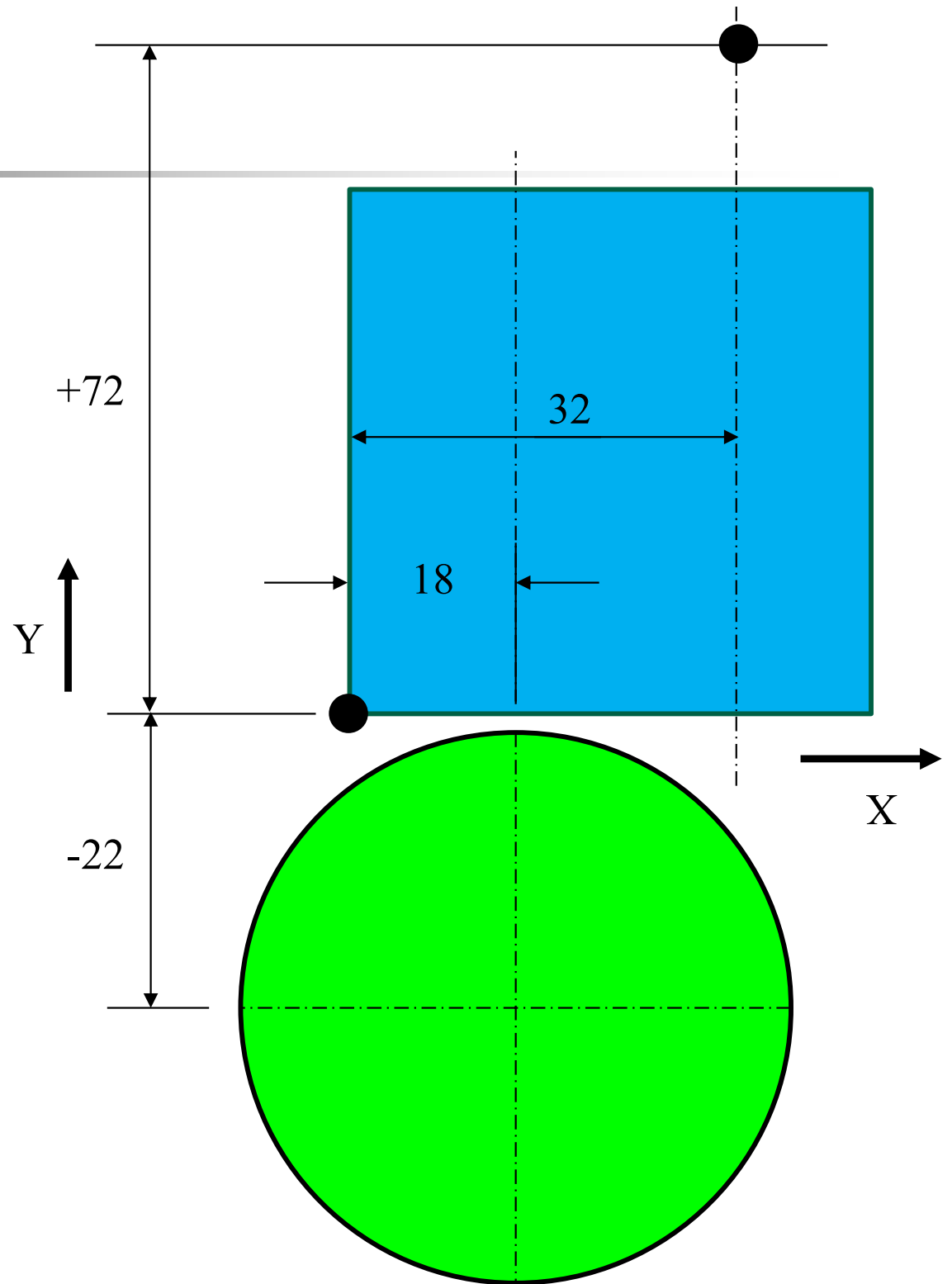
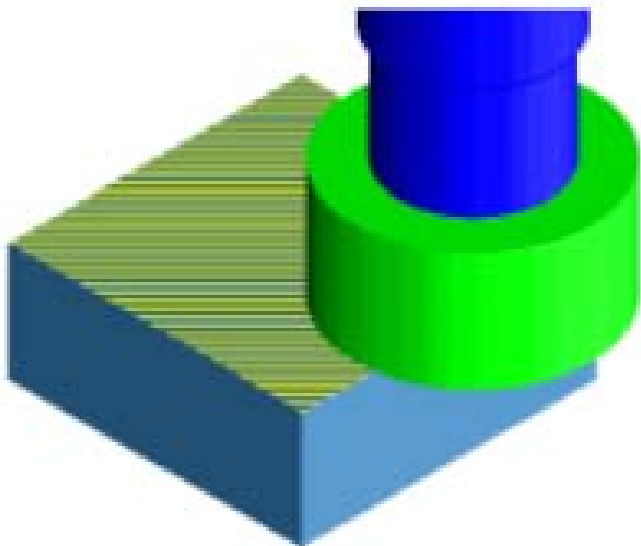
Step 1: Facing

Starting the program:

- **N05 G90 G17 G71 G40**
- **N10 G54**
- **N15 T1 D1 M6 ; SHELL ENDMILL Ø40**
- **N20 S1000 G94 F100 M3 M8**

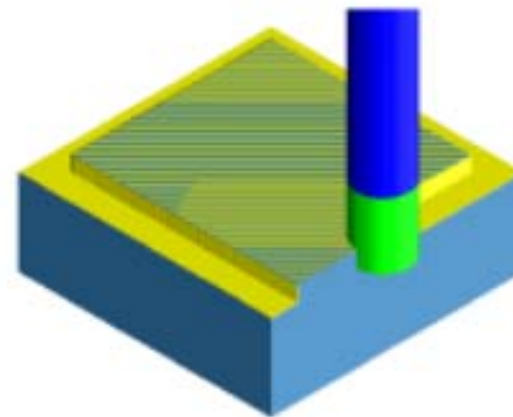
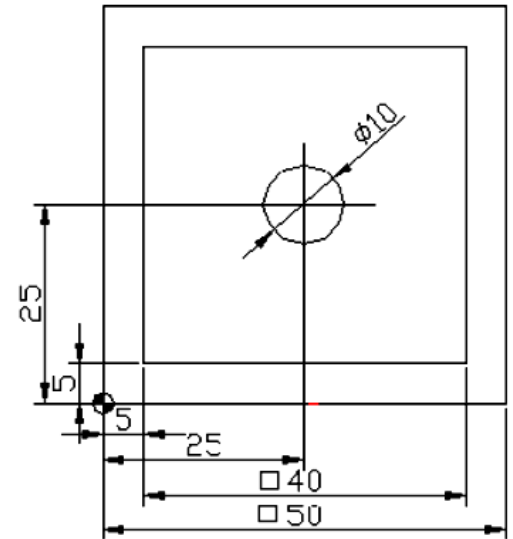
Step 1: Facing

- N25 G0 X18 Y-22
- N30 Z0
- N35 G1 Y72
- N40 G0 X32
- N45 G1 Y-22
- N50 G0 Z10



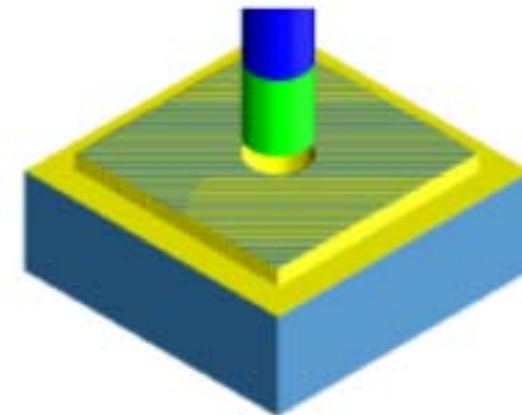
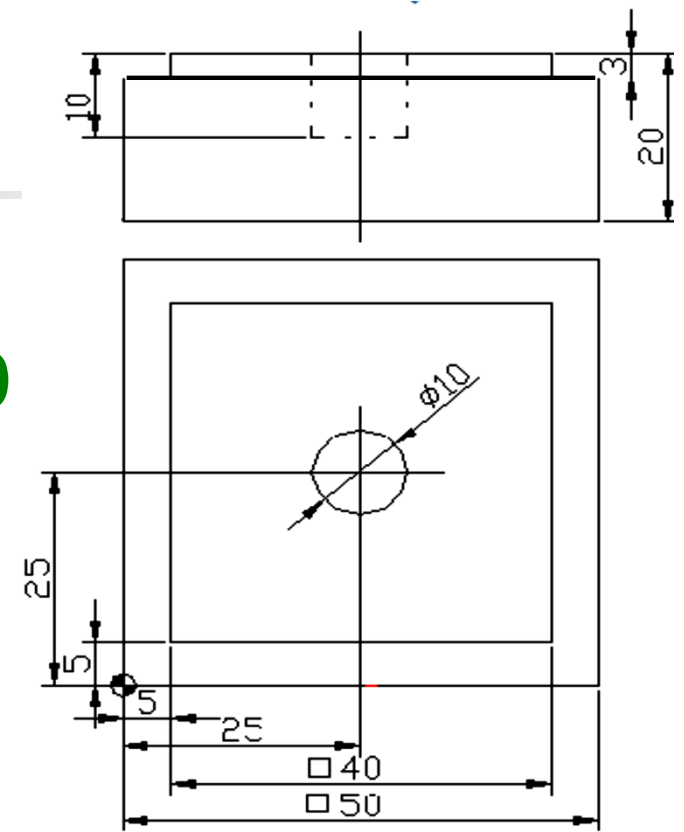
Step 2: End Milling

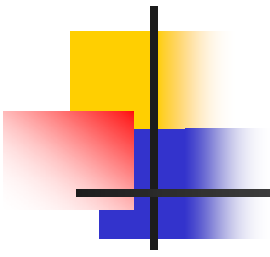
- **N55 T2 D1 M6 ; SHANK ENDMILL Ø16**
- **N60 S1500 G94 F150 M3 M8**
- **N 65 G0 X-3 Y-10**
- **N 70 Z2**
- **N 75 G1 Z-3**
- **N80 Y53**
- **N85 X53**
- **N90 Y-3**
- **N95 X-10**
- **N100 G0 Z10**



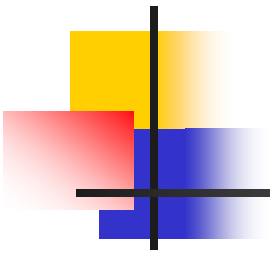
Step 3: Slot Drilling

- N110 T3 D1 M6 ; **SLOT DRILL Ø10**
- N120 S2000 G94 F200 M3 M8
- N130 G0 X25 Y25
- N140 Z2
- N150 G1 Z-10
- N160 G0 Z90
- N170 M5 M9
- N180 M30 ; **END PROGRAM**





QUESTIONS???



END